~~52010-XX~~

~~Tryckspecifikation~~

| | |
|---|---|
| Format: | 420 x 210 mm |
| Färger: | PMS svart |
| Utskrift: | anpassat till A3 |
| Pharm. code: 1864 | |

~~5:e korr 20-09-04~~
~~Krzys Liberda, 018-64 43 69~~

# *Intralipid® 20%*

## Emulsion for infusion

Trade name of the medicinal product
Intralipid20%, Emulsion forinfusion

Qualitative and quantitative composition

Intralipid 20%
1000 ml of the emulsion contains: Purified soybean oil 200 g, purified egg phospholipids, glycerol anhydrous, sodium hydroxide,waterforinjections
pH appr 8
Osmolality: 350 mosm/kg water
Energycontent: 8.4 MJ (2000 kcal) / 1000 ml
Organicphosphate content: 15 mmol / 1000 ml

PHARMACEUTICAL FORM

Emulsion for infusion.
Whitehomogenous emulsion.

CLINICAL PARTICULARS

**Therapeutic indications**

INTRALIPID is indicated in patients needing intravenous nutrition to supply energy and essential fatty acids. INTRALIPID is also indicated in patients with essential fatty acid deficiency(EFAD) who cannot maintain or restore a normal essential fatty acid pattern by oral intake.

**Posology and method of administration**

Theabilitytoeliminate INTRALIPIDshouldgovernthedosageand infusion rate. See Fat elimination.

Adults: The recommended maximum dosage is 3 gm triglycerides/kg body weight / day. Intralipid can supply up to 70% of the energy requirements, also in patients with highly increased energy requirements. The infusion rate for Intralipid 10% and 20% should not exceed 500 ml in 5 hours.

Neonates and infants:
The recommended dosage range in neonates and infants is 0.5 - 4 gm triglycerides/kg/ day. The rate of infusion should not exceed 0.17 gm triglycerides/kg/hour (4g/kg in 24 hours). In premature and low birth weight neonates, Intralipid should preferably be infused continuously over 24 hours. The initial dosage of 0.5 - 1 gm/kg/day up to 2 gm/kg/day.

Only with close monitoring of serum TG concentration, liver test and oxygen saturation may the dosage be increased to 4 gm/kg/day. No attemp should be made to exceed these rates in order to compensate for missed doses.

**Essential fatty acid deficiency.**
To prevent or correct essential fatty acid deficiency, 4 - 8% of the nonprotein energy should be supplied as Intralipid to provide sufficient amounts of linoleic and linoleic acids. When EFAD is associated with stress, the amount of Intralipid needed to correct the deficiency may be substantially increased.

**Contra-indications**
-Hypersensitivity to the active substances or any of the excipients listed.

-Hypersensitivity to egg protein.

-Hypersensitivity to soya- or peanut protein. Intralipid contains soybean oil.

Cross allergic reactions have been observed between soybean and peanut.

- Acute shock.

- Conditions involving severe hyperlipidaemia.

-Severe liver failure.

-Haemophagocytic syndrome.

**Special warnings and special precautions for use**
Caution should be exercised in conditions with impaired lipid metabolism such as renal insufficiency, uncontrolled diabetes, pancreatitis, hepatic insufficiency, hypothyroidism (if hypertriglyceridaemia is present) and sepsis. If Intralipid is administered in these conditions, the serum concentration of triglycerides must be closely monitored.

Intralipid may interfere with certain analyses (bilirubin, lactate dehydrogenase, oxygen saturation, Hb etc.) if the blood sample is taken before the administered fat has been eliminated from the blood. In most patients, the fat is eliminated from the blood 5-6 hours after infusion has been completed.

Paediatric population

Intralipid must be administered with caution to infants and premature babies with hyperbilirubinemia and in cases of suspected pulmonary hypertension.

Light exposure of solutions for intravenous parenteral nutrition, especially after admixture with trace elements and/or vitamins, may have adverse effects on clinical outcome in neonates, due to generation of peroxides and other degradation products. When used in neonates and children below 2 years, Intralipid should be protected from ambient light until administration is completed (see sections Posology, Instruction of Use).

This medicine contains less than 1 mmol sodium (23 mg) per dosage unit, that is to say essentially 'sodium-free'.

**InteractionswithOtherMedicaments**
Certain medicinal products such as insulin can affect the body's lipase system. However, this type of interaction appears to have limited clinical significance.

Heparin in clinical doses causes transient increase of plasma lipolysis resulting in a temporary reduction of triglyceride clearance due to reduced lipoprotein lipase.

Soybean oil has a natural content of vitamin $K_1$. This is considered important only in patients treated with coumarin derivatives, which interact with vitamin $K_1$.

**Pregnancy and lactation**

Data from a limited number of pregnancies indicate no harmful effects of Intralipid on pregnancy or fetal/neonatal health. Other epidemiological data of significance are lacking. Intralipid should be prescribed with caution to pregnant women.

**Effects on ability to drive and use machines**

Intralipid has no or negligible influence on the ability to drive and use machines.

**Undesirable effects**

| | Uncommon<br>≥1/1000, <1/100 | Rare<br>≥1/10 000, <1/1000 |
|---|---|---|
| General disorders and administration site conditions | Fever, shivering, nausea. | Anaphylactic reaction, headache, abdominal pain, fatigue |
| Respiratory, thoracic and mediastinal disorders | | Tachypnoea |
| Blood and lymphatic system disorders | | Haemolysis, reticulocytosis, thrombocytopenia (in conjunction with long-term treatment of infants) |
| Vascular disorders | | Hyper/hypotension |
| Skin and subcutaneous tissue disorders | Flush | Exanthema, urticaria |
| Hepatobiliary disorders | | Liver impairment |
| Reproductive system and breast disorders | | Priapism |

Reports of undesirable effects other than fever, sensation of heat, shivering and nausea in connection with the administration of Intralipid are very uncommon (less than one report per million infusions). The rare undesirable effects can occur during or after infusion (early effects) or after long-term treatment. Early effects that have occurred at recommended dosage include allergic reactions, abdominal pain, haemolysis, headache, hyper/hypotension, priapism, reticulocytosis, tachypnoea and fatigue.

Effects reported after long-term treatment include thrombocytopenia in infants.

***"Fat overload syndrome"***

Impaired ability in the patient to eliminate Intralipid can lead to "Fat overload syndrome" as a result of overdose. This syndrome can occur in cases of pronounced hypertriglyceridaemia and may also arise at the recommended infusion rate in connection with a sudden change in the patient's condition, such as deterioration of renal function or infection.

"Fat overload syndrome" is characterised by hyperlipidaemia, fever, fat infiltration, hepatomegaly, splenomegaly, anaemia, leukopenia, thrombocytopenia, coagulation disturbance and coma. The symptoms usually disappear if treatment is discontinued.

Transient changes in liver function tests have been found in conjunction with intravenous nutrition but the cause is unknown.

**Overdose**

See Undesirable effects, "Fat overload syndrome". Marked overdose of fat emulsion containing triglycerides may lead to metabolic acidosis, particularly if carbohydrates are not administered concomitantly.

**PHARMACOLOGICAL PROPERTIES**

Pharmacotherapeutic group: Solutions for parenteral nutrition, fat emulsions. ATC code:

B05BA02.

Intralipid is a fat emulsion for intravenous use containing soybean oil emulsified with egg phospholipids. Particle size and biological properties are similar to those of natural chylomicrons. Unlike chylomicrons, Intralipid does not contain cholesterol esters or apolipoproteins while its phospholipid content is considerably higher.

**Pharmacokinetic properties**

Fat particles in Intralipid are essentially distributed and eliminated the same way as natural chylomicrons.

PHARMACEUTICALPARTICULARS

**Incompatibilities**

INTRALIPID can only be mixed with other medicinal products for which compatibility has been documented.

**Storage Condition**

Storebelow 25°C.Donotfreeze.

*Infusion bottle.* Type II glass and butyl rubber stopper.

Pack sizes:
Intralipid20% 100 ml, 250 ml, 500ml

**Instructions for use/handling**

Do not use if package is damaged.
Additions should be made aseptically. Single administration of electroly te solutions to INTRALIPID should not be made. Only medicinal, nutri tional or electrolyte solutions for which compatibility has been docume nted may be added as directed. Compatibility data are available from the manufacturer for a number of mixtures. The left over contents of opened bottles / bags should be discarded and not saved for later use.

**DATE OF REVISION OF THE TEXT**

October2021

MY Version: PI835025-3700

Manufactured by Fresenius Kabi SSPC, Wuxi, China for

Fresenius Kabi AB, 751 74 Uppsala, Sweden